

PAPER-1

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

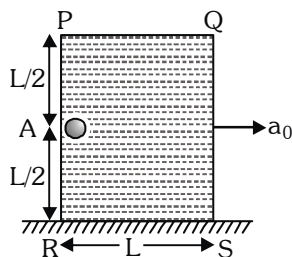
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

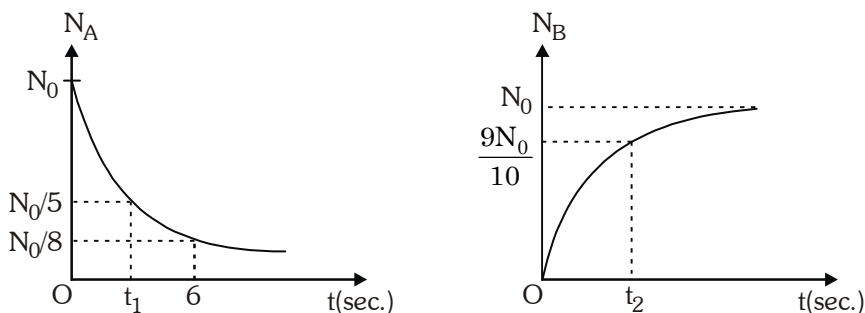
1. A small solid ball of density ρ_0 is held inside at point A in a closed cubical container of side L, filled with an ideal liquid of density $4\rho_0$ as shown in the figure. Now, if the container starts moving with constant acceleration a_0 horizontally and the ball is released from point A simultaneously, It has been observed that the ball hits the point Q after time of T_0 . Points A, P, Q, R and S are lying in the same plane. Then choose the **CORRECT** option(s) :



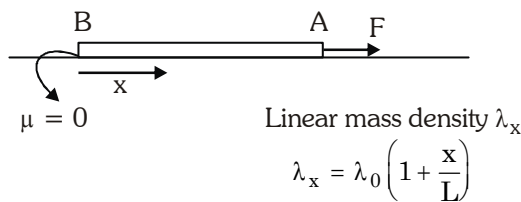
- (A) For ball to hit the top of container at end Q, $a_0 = \frac{3g}{2}$
 (B) For ball to hit the top of container at end Q, $a_0 = 2g$
 (C) Ball hits the top of container at end Q after a time $T_0 = \sqrt{\frac{L}{3g}}$
 (D) Ball hits the top of container at end Q after a time $T_0 = \sqrt{\frac{2L}{3g}}$

Space for Rough Work

2. In a radioactive decay process A decays to B, $A \rightarrow B$. Two graphs of number of nuclei of A and B versus time is given. Then choose the **CORRECT** option(s)



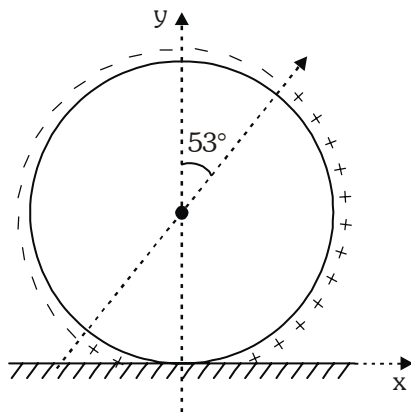
- (A) $t_2 - t_1 = 4$ (B) $t_2 - t_1 = 2$ (C) $t_1 = 2 \log_2 5$ (D) $t_2 = \log_2 100$
3. A non-uniform rope of length L is placed on the ground and Rope is pulled by constant force F_0 as shown in the diagram, there is no friction anywhere, linear mass density of rope is given by $\lambda_x = \lambda_0 \left(1 + \frac{x}{L}\right)$, where x is measured from the end B. If tension at a distance of x from B is T_x , total mass of the rope is M and acceleration of rope is a_0 . Then choose the **CORRECT** option(s).



- (A) $\int_0^L T_x \cdot dx = \frac{4F_0 L}{9}$ (B) $M = \frac{3\lambda_0 L}{2}$ (C) $a_0 = \frac{2F_0}{3\lambda_0 L}$ (D) $T_x = \frac{2F_0}{3L} \left[x + \frac{x^2}{2L} \right]$

Space for Rough Work

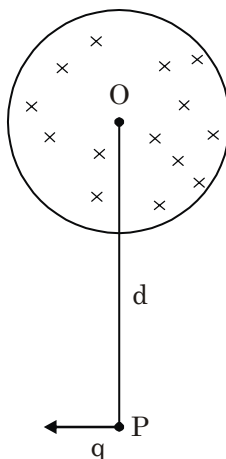
4. A non conducting uniform thin shell of mass m and radius R is having uniform charge density $+\sigma$ on one half and $-\sigma$ on another half as shown in figure. It is placed on a rough non conducting horizontal plane. At $t = 0$ a uniform electric field $\vec{E} = -E_0 \hat{j} \text{ N/C}$ is switched on and the solid sphere starts rolling without slipping. Then which of the following is/are **CORRECT**.



- (A) The acceleration of the topmost point of the hollow sphere at $t = 0$, is $\frac{36}{25} \frac{\pi \sigma E_0 R^2}{m}$.
- (B) The magnitude of frictional force at $t = 0$ is $\frac{18}{25} \pi \sigma E_0 R^2$.
- (C) The acceleration of centre of mass of the hollow sphere at $t = 0$, is $\frac{12}{25} \frac{\pi \sigma E_0 R^2}{m}$.
- (D) The magnitude of frictional force at $t = 0$ is $\frac{9}{25} \pi \sigma E_0 R^2$.

Space for Rough Work

5. A point object is placed at 30 cm from a convex glass lens $\left(\mu_g = \frac{3}{2}\right)$ of focal length 20 cm. The final image of object will be formed at infinity if -
- (A) Another concave lens of focal length 60 cm is placed in contact with the previous lens
- (B) Another convex lens of focal length 60 cm is placed at a distance of 30 cm from the first lens
- (C) The whole system is immersed in a liquid of refractive index $4/3$
- (D) The whole system is immersed in a liquid of refractive index $9/8$
6. In the cylindrical region of radius $R = 2\text{m}$, there exists a time varying magnetic field B such that $\frac{dB}{dt} = 2 \text{ Tesla/sec}$. A charged particle having charge $q = 2\text{C}$ is placed at the point P at a distance d from its center O. Now, the particle is moved in the direction perpendicular to the line OP by an external agent up to infinity so that there is no gain in kinetic energy of charged particle. If $(d > R)$, then which of the following options is/are **CORRECT**?



- (A) work done by external agent is 4π Joule when $d = 4\text{m}$
- (B) work done by external agent is 4π Joule when $d = 8\text{m}$.
- (C) work done by external agent is independent of d .
- (D) work done by external agent is positive.

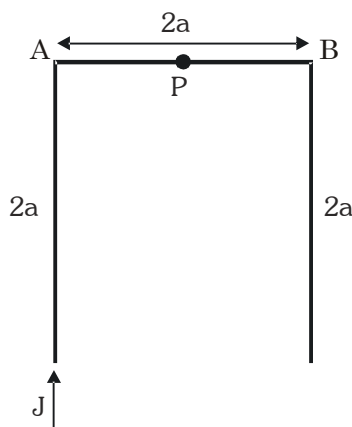
Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

A uniform wire frame of linear mass density λ having three sides each of length $2a$ is kept on a smooth horizontal surface. An impulse J is applied at one end as shown in the figure. P is the midpoint of AB.

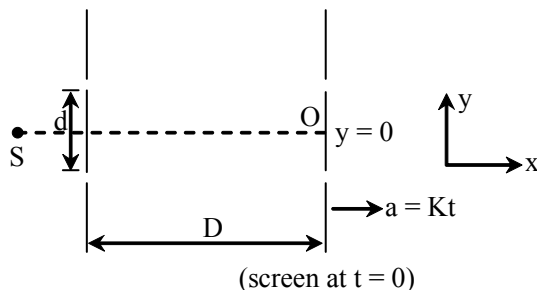


7. The angular velocity of system just after the impulse is
- (A) $\frac{3J}{22\lambda a^2}$ (B) $\frac{3J}{11\lambda a^2}$ (C) $\frac{J}{11\lambda a^2}$ (D) $\frac{2J}{11\lambda a^2}$
8. The velocity of point P just after the impulse is
- (A) $\frac{J}{\lambda a} \hat{j}$ (B) $\frac{J}{6\lambda a} \hat{j}$ (C) $\frac{J}{\lambda a} \left(\frac{2}{11} \hat{i} + \frac{1}{6} \hat{j} \right)$ (D) $\frac{J}{\lambda a} \left(\frac{1}{11} \hat{i} + \frac{1}{6} \hat{j} \right)$

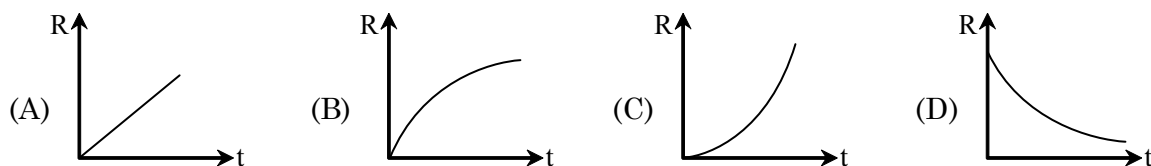
Space for Rough Work

Paragraph for Questions 9 and 10

Consider YDSE arrangement shown. The screen in the arrangement starts accelerating from rest in the positive x-direction with acceleration $a = Kt$ at time $t = 0$ (here K is a constant and t is time)



9. Rate of change of fringe width (R) for the fringe pattern on the screen varies with time as



10. Velocity of n^{th} maxima (above "O") at any time t in ground frame is -

(A) $\frac{Kt^2}{2} \left(\hat{i} + \frac{n\lambda}{d} \hat{j} \right)$ (B) $Kt \left(\hat{i} + \frac{n\lambda}{d} \hat{j} \right)$ (C) $\frac{Kt^2}{2} \left(\frac{n\lambda}{d} \hat{i} + \hat{j} \right)$ (D) None of these

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

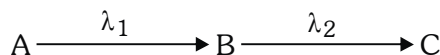
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+	●
●	●
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9

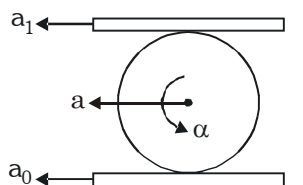
●	-
●	●
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- In a radioactive decay A is converting into B and B is converting into C. It is given that at $t = 0$ nuclei of B and C are not present, decay constant for A to B is λ_1 and decay constant for B to C is λ_2 , take $\lambda_1 = 5\lambda$ and $\lambda_2 = 2\lambda$. The time after which B will have maximum No. of nuclei is of the form $\frac{\ln(5/2)}{2B_0\lambda}$. The value of B_0 will be :-



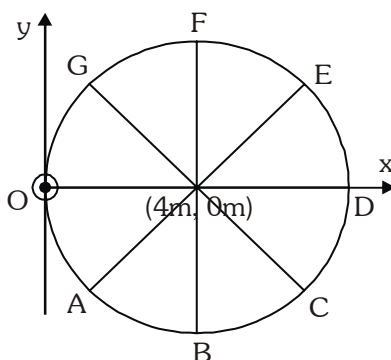
- A system of two planks and a sphere of radius R is in motion as shown in figure. Radius of the sphere is R and there is no slipping anywhere. It is given that $R\alpha = 2a_0$ where α is angular acceleration of sphere and acceleration of upper block is $a_1 = ka_0$ where k is a +ve constant then the value of k will be :



Space for Rough Work

3. An infinite uniform current carrying wire is kept along z-axis, carrying current I_0 in the direction of the positive z-axis. OABCDEFG represents a circle (where all the points are equally spaced) whose centre at point $(4m, 0m)$ and radius $4m$ as shown in the figure.

$$\int_{DEF} \vec{B} \cdot d\vec{\ell} = \frac{\mu_0 I_0 k}{2} \text{ in S.I. unit, then the value of } k \text{ is :}$$



4. A closed organ pipe of length L and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases are equal in both the pipes. Both the pipes are

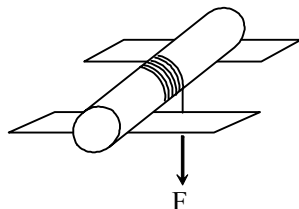
vibrating in their first overtone with same frequency. The length of open organ pipe is $kL \left(\frac{\rho_1}{\rho_2} \right)^{1/2}$.

Find value of k

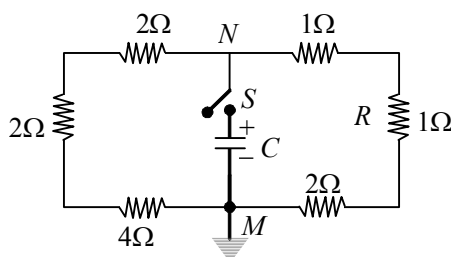
5. A non-conducting thin disc of radius R charged uniformly over one side rotates uniformly with an angular velocity ω about an axis passing through its periphery point P and perpendicular to its plane. The electrostatic potential at the point P due to charges on the disc is V_0 before the rotation. The magnetic field at P due to the moving charges upon rotation is $2k\mu_0 \epsilon_0 \omega V_0$. Find the value of k

Space for Rough Work

6. A uniform solid cylinder of mass 'm' rests on two horizontal planks. A thread is wound on the cylinder. The hanging end of the thread is pulled vertically down with a constant force F , which still does not bring any sliding of the cylinder. If the coefficient of friction between the cylinder and planks is equal to μ . What is the maximum acceleration $a_{\max}$ of the axis of cylinder rolling over the planks. (Assume $\mu = 1/3$ and take $g = 10 \text{ m/sec}^2$)



7. A capacitor of capacity $6 \mu\text{F}$ and initial charge $160 \mu\text{C}$ is connected with a key S and resistance as shown in figure. Point M is earthed. If key is closed at $t = 0$ then the current through resistance $R = 1 \Omega$ at $t = 16 \mu\text{sec}$ will be ___ Amp. (Assume $e = 2.7$)



8. The distance between the parallel plates of a isolated charged condenser is $d = 5 \text{ cm}$ and the intensity of the field $E = 300 \text{ V cm}^{-1}$. A slab of dielectric constant $K = 5$ and 1 cm wide is inserted parallel to the plates. Determine the potential difference (in volt) between the plates, after the slab is inserted.

Space for Rough Work

PART-2 : CHEMISTRY

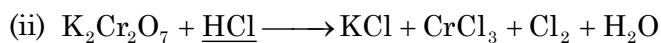
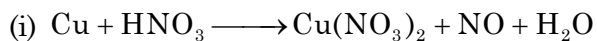
SECTION-I(i) : (Maximum Marks: 24)

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-
1. Select the incorrect statement(s) among given
 (A) The highest occupied molecular orbital (HOMO) in N_2 has no s-character.
 (B) Anhydrous $CaCl_2$ can be used for drying of NH_3 .
 (C) Pt can form more stable complex with +4 oxidation state as compare to Ni^{+4} .
 (D) $(O_2)[PF_6]$ is paramagnetic.
 2. Find correct statements for Coordination compounds.
 (A) $[PtCl_2(NH_3)_2]$ & $[Cu(NH_3)_4]^{2+}$ are square planar complexes.
 (B) Complex $[Fe(C_2O_4)_3]^{3-}$, does not exhibit geometrical isomerism while exhibits optical isomerism.
 (C) CO bond order : $[Ni(CO)_4] > [Co(CO)_4]^- > [Fe(CO)_4]^{2-}$.
 (D) $[Ni(Cl_2)(PPh_3)_2]$ has two geometrical isomers.
-

Space for Rough Work

3. Equivalent weight of the underlined (HNO_3 as oxidising agent and HCl as reducing agent) substances in the given reaction is respectively (approx).



(A) 63, 36.5

(B) 63, 73

(C) 84, 85

(D) 84, 42.5

4. The radial distribution functions $[P(r)]$ is used to determine the most probable radius, which is used to find the electron in a given orbital $\frac{dP(r)}{dr}$ for 1s-orbital of hydrogen like atom having atomic number Z , is

$$\frac{dP(r)}{dr} = \frac{16\pi Z^3}{a_0^3} \left(2r - \frac{2Zr^2}{a_0} \right) e^{-2Zr/a_0}$$

Which of the following statement(s) is/are correct ?

- (A) At the point of maximum value of radial distribution function, $\frac{dP(r)}{dr} = 0$; one antinode is present
- (B) Most probable radius of Li^{2+} is $\frac{a_0}{3}$
- (C) Most probable radius of He^+ is $\frac{a_0}{2}$
- (D) Most probable radius of hydrogen atom is a_0

Space for Rough Work

5. Consider the following table and select correct option/s.

Compound	bond dissociation energy(kJ/mole)	Bond angle	Boiling point (°C)
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H ₂ O	a ₁	b ₁	c ₁
H ₂ S	a ₂	b ₂	c ₂
H ₂ Se	a ₃	b ₃	c ₃
H ₂ Te	a ₄	b ₄	c ₄

- (A) $a_1 > a_2 > a_3 > a_4$ (B) $b_1 > b_2 > b_3 > b_4$ (C) $c_1 > c_2 > c_3 > c_4$ (D) $c_2 < c_3 < c_4 < c_1$

6. How many of the following statement is/are **INCORRECT**?

- (A) EAN of [Rh(ox)₃]³⁻ is 56
 (B) d_{C-C} in Zeise's salt is less than that of C₂F₄ molecule
 (C) Red bauxite consists of Fe₂O₃ as minor impurity
 (D) In Hall-Heroult process, the produced Al floats at the top of the electrolyte

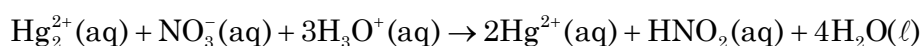
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SECTION-I(ii) : (Maximum Marks : 12)

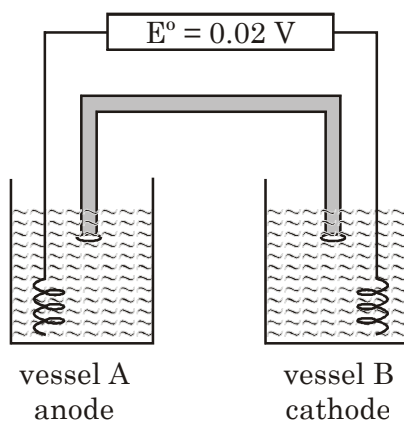
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Paragraph for Questions 7 and 8

The cell potential for the chemical reaction :



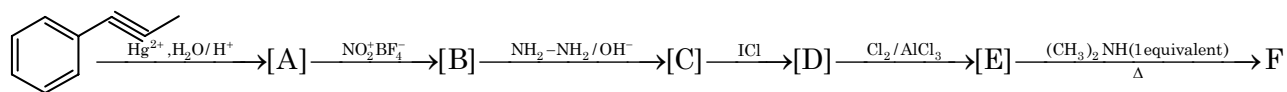
is measured under standard conditions in the electrochemical cell shown in the accompanying diagram.



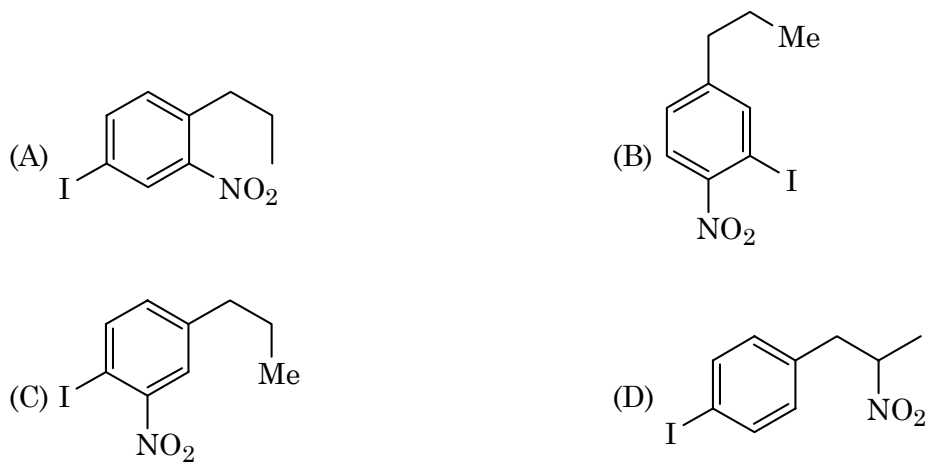
7. How many moles of electron pass through the circuit when 0.60 mole of Hg_2^{2+} and 0.30 mole of HNO_2 are produced in the cell that contain 0.50 mole of Hg_2^{2+} and 0.40 mole of NO_3^- at the beginning of the reaction ?
 (A) 0.30 (B) 0.60 (C) 0.15 (D) 1.20
8. How long will it take to produce 0.10 mole of HNO_2 by this reaction if the cell delivers a constant current of 10 A ?
 (A) 965 s (B) 193 s (C) 1930 s (D) 482.5 s

Space for Rough Work

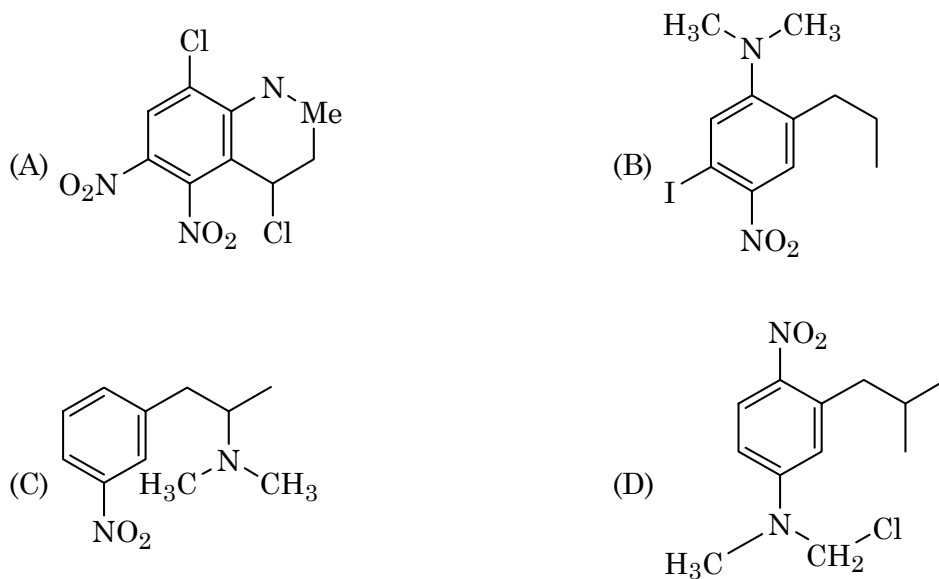
Paragraph for Questions 9 and 10



9. The compound (D) is



10. The compound F is



Space for Rough Work

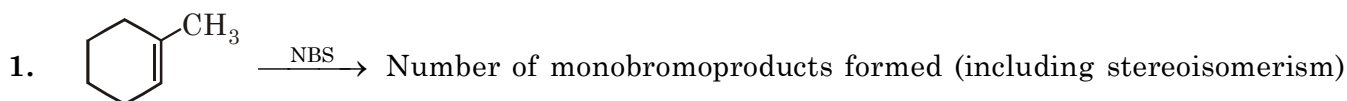
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For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+						-					
●	●	0	0	0	0	●	●	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	●	2	2	2	2	2	●	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	●	5	5	5	●	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.



2. Following data is given for two ideal gas samples taken in two separate containers :

Gas	Pressure	Volume	Temperature	Mean free paths(λ)
X	2 atm	V litre	800 K	0.08 nm
Y	8 atm	V litre	200 K	0.02 nm
&	$\frac{(Z_{11})_X}{(Z_{11})_Y} = \frac{1}{16}$			

Find the ratio of rate of diffusion of gas X and gas Y in similar condition of pressure, temperature, etc.

Space for Rough Work

3. When P_4 reacts with NaOH & gives a gas (A) & forms another compound (B) containing phosphorous. Find out the difference in oxidation state of Phosphorous in (A) and (B).

(Consider A and B as a major product)

4. (i) $Ti^{+3} \Rightarrow$ Purple (ii) $Fe^{+2} \Rightarrow$ Green
 (iii) $Fe^{+3} \Rightarrow$ Yellow (iv) $Ni^{+2} \Rightarrow$ Green
 (v) $Cu^{+2} \Rightarrow$ Blue (vi) $Sc^{3+} \Rightarrow$ Colourless
 (vii) $Ti^{4+} \Rightarrow$ Colourless (viii) $Zn^{2+} \Rightarrow$ Colourless

How many colour of aqueous cation are correctly matched.

5. In the saturated aqueous solution of $PbCl_2$, the freezing point of water decreases by $\left(\frac{D}{100}\right)^\circ C$,

then 'D' is :

(Given k_{sp} of $PbCl_2 = 4 \times 10^{-6}$, $K_{f, water} = 2 \text{ K kg/mole}$, assume molality = molarity)

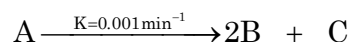
6. A reversible cyclic process involves 6 steps. In step -1 and 3, system absorb 500 J and 800 J of heat from a heat reservoir at temperature 250K & 200K, respectively. Step 2, 4, 6, are adiabatic such that the temperature of one reservoir changes to that of next. Total work done by the system in whole cycle is 700 J. Find the temperature (in K) during step 5 if it exchange heat from a reservoir at temperature T_5

Space for Rough Work

7. Various isomeric alkynes have molecular formula $C_3FClBrI$. For such alkynes if
 p = Number of positional isomers
 q = Number of optically active isomers
 r = Number of fractions obtained on fractional distillation of a mixture of all possible alkynes
 s = Number of diastereomeric pairs

Then find the value of $\left(\frac{p+q-r+s}{5}\right)$.

8. An optically active substance 'A' decomposes into optically active substances 'B' and 'C' as follows



The specific rotations of A, B and C are $+40^\circ$, $+10^\circ$ and -30° per mole, respectively. If initially A and C were present in 4 : 3 mole ratio, the time (in min), after which the sample becomes optically inactive is ($\ln 2 = 0.7$, $\ln 5 = 1.6$, $\ln 7 = 2.0$, $\ln 13 = 2.5$)

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Consider two sets

$$A = \left\{ \frac{p}{q}; p, q \text{ are coprime odd integers} \right\} \text{ and}$$

$$B = \left\{ \frac{p+1}{q}; p, q \text{ are odd integers and } p \neq -1 \right\}.$$

Functions $f : A \rightarrow \mathbb{R}$ & $g : B \rightarrow \mathbb{R}$ are both defined as $f(x) = x^x$ & $g(x) = x^{|x|}$ on respective domains. Then :

(A) $f\left(\frac{1}{3}\right) \cdot g\left(\frac{-2}{3}\right) = \frac{\sqrt[3]{4}}{3}$

(B) $g(x)$ is an even function.

(C) $f(1) + g(2) = 5$

(D) $\left[g\left(\frac{-6}{5}\right) \right]^{5/2} - \left[g\left(\frac{4}{5}\right) \right]^{15/4} - \left[f\left(\frac{3}{5}\right) \right]^5 = 1$

Space for Rough Work

2. If sides of triangle ABC are a, b and c such that $2b = a + c$ then
 (A) $\frac{b}{c} > \frac{2}{3}$ (B) $\frac{b}{c} > 2$ (C) $\frac{b}{c} < 2$ (D) $\frac{b}{c} < \frac{3}{2}$
3. If z_1, z_2, z_3 are any three roots of the equation $z^6 = (z + 1)^6$, then $\arg\left(\frac{z_1 - z_2}{z_2 - z_3}\right)$ can be equal to
 (A) 0 (B) π (C) $\frac{\pi}{4}$ (D) $-\frac{\pi}{4}$
4. If foci of an ellipse be $(-1, 2)$ and $(-2, 3)$ and its tangent at a point A is $2x + 3y + 9 = 0$
 (A) Length of the minor axis of the ellipse will be $2\sqrt{14}$
 (B) Co-ordinate of the point 'A' will be $\left(\frac{-32}{9}, \frac{-17}{27}\right)$
 (C) Distance between the foci is $2\sqrt{2}$
 (D) Product of the perpendiculars from foci to any tangent is 56
5. Let A_1, A_2, A_3, A_4 are in G.P. If the H.M. of A_1, A_2 is 12 and that of A_2, A_3 is 36, then
 (A) $3A_1 - A_2 = 4$ (B) $9A_1 - A_3 = 0$
 (C) $A_4 - 4A_2 - 2A_3 = -24$ (D) The sum of digits in A_4 is 6
6. Let the function $f(x) = \begin{cases} 2x - x^3 + \ln(a^2 - 3a + 3), & 0 \leq x < 2 \\ x - 6, & x \geq 2 \end{cases}$. It possess a local minimum at $x = 2$ for
 (A) $a = 0$ (B) $a = 3$ (C) $a = 4$ (D) $a = -1$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
 - Based on each paragraph, there are **TWO** questions.
 - Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
 - For each question, darken the bubble corresponding to the correct option in the ORS.
 - For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases
-

Paragraph for Questions 7 and 8

Consider the quadratic equation $2x^2 - (4m + 2)x + m^2 + m = 0$, $m \in \mathbb{R}$

7. The number of positive integer values of 'm' such that the equation has exactly one root in (2, 3) is
(A) 3 (B) 4 (C) 5 (D) 6
8. The number of negative integral values of 'm' such that $m > -10$ and atleast one root of the equation is smaller than '2' is
(A) 8 (B) 9 (C) 6 (D) 4
-

Space for Rough Work

Paragraph for Questions 9 and 10

Consider a standard ellipse whose focus and corresponding foot of directrix are $(\sqrt{7}, 0)$ and $(\frac{16}{\sqrt{7}}, 0)$ and a circle with equation $x^2 + y^2 = r^2$. If in the first quadrant, the common tangent to a circle of this family and the above ellipse meets the co-ordinate axes in A and B then

9. If 'P' is a variable point on the above ellipse with S, S' are foci then the maximum possible area of $\Delta PSS'$ is

(A) $\sqrt{7}$ sq. units (B) $2\sqrt{7}$ sq. units (C) $3\sqrt{7}$ sq. units (D) $4\sqrt{7}$ sq. units

10. Slope of line joining AB when $r = \frac{7}{2}$ is

(A) $-\sqrt{\frac{13}{15}}$ (B) $\sqrt{\frac{13}{15}}$ (C) $\sqrt{\frac{15}{13}}$ (D) $-\sqrt{\frac{15}{13}}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+		-	
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Given : $S_n = \sum_{r=1}^n t_r = \frac{n(2n^2 + 9n + 13)}{6}$ Find $\sum_{r=1}^{\infty} \frac{1}{r \cdot \sqrt{t_r}}$
- Let $I(n) = \int_1^e x^3 \cdot (\ln x)^n dx$, where $n \in \mathbb{N}$. Find the value of $\ell n(4I(5) + 5I(4))$.
- Consider a sequence $f_n(x)$ defined as $f_1(x) = 1$ & $f_{n+1}(x) = \int_0^{\infty} \frac{e^{-tx}}{f_n(t)} dt$. Evaluate the value of $\frac{f_8(2)}{f_6(2)}$.

Space for Rough Work

4. Let $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$. If $A^3 - 5A^2 + 9A = B$. Find trace of matrix B.
5. If $f(x) = \frac{ax+b}{bx-a}$, $x \neq \frac{a}{b}$, $a > 0$, $b > 0$ and if $g(x) = f(f(x)) + f\left(f\left(\frac{1}{x}\right)\right)$ and $g : [1, \infty) \rightarrow [2, \infty)$ then the value of x for which $g(x) = \frac{5}{2}$, is
6. It is given that $\log_6 a + \log_6 b + \log_6 c = 6$, where a, b and c are positive integers that form an increasing geometric sequence and $b - a$ is the square of an integer. Find $a + b + c$.

Space for Rough Work

7. Evaluate the value of the limit $L = \lim_{x \rightarrow 0} \left(\frac{1}{\ln(x + \sqrt{x^2 + 1})} - \frac{1}{\ln(x + 1)} \right)$
8. Consider the curves $C_1 : |z - 2| = 2 + \operatorname{Re}(z)$ and $C_2 : |z| = 3$ (where $x, y \in \mathbb{R}$ and $i = \sqrt{-1}$). They intersect at P and Q is the first and fourth quadrants respectively. Tangents to C_1 at P and Q intersect the x-axis at R and tangents to C_2 at P and Q intersect the x-axis at S. If area of ΔPRS is $\lambda\sqrt{2}$ sq.units, then the value of (λ^2) .

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.
17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator